Sound Becomes a Signal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Sound Becomes a Signal before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Sound Becomes a Signal trace and record the first step you can explain without help.

Use W2: Inspect what was discarded for Sound as **sample**d information. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Sound Becomes a Signal explanation against the supplied facts before

> discussing it.

Use the worked case for Sound as **sample**d information. Proposed explanation: “Five **sample**s make five gaps”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Construct a new hidden change

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Sound Becomes a Signal target to a trace and a justified conclusion.

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Sound Becomes a Signal case and show the deciding evidence.

A microphone records quiet and loud claps at the same distance and settings.

> What **waveform** difference is expected? Does it reveal a spoken word?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 6 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Sound Becomes a Signal adds to Helpful Classroom Sorter and keep the evidence.

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Read samples and gaps

> Read samples and gaps. Apply the supplied Sound Becomes a Signal facts and keep your reasoning.

Five **sample**s have values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. State the **sample** count, interval and elapsed time from first to last. Explain why there are four gaps.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Inspect what was discarded

> Inspect what was discarded. Apply the supplied Sound Becomes a Signal facts and keep your reasoning.

Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original **signal** that these retained values do not justify.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Compare equal summaries

> Compare equal summaries. Apply the supplied Sound Becomes a Signal facts and keep your reasoning.

Sequences A=0,2,0,2,0 and B=0,0,2,0,0 use the same time interval. State their maximum values and one difference the maximum discards. Name information that would preserve that difference.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Construct a new hidden change

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Construct a new hidden change. Apply the supplied Sound Becomes a Signal facts and keep your reasoning.

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

My reasoning and evidence

W5 Check a speech pipeline

> Check a speech pipeline. Apply the supplied Sound Becomes a Signal facts and keep your reasoning.

A recording says Set the timer for fourteen minutes. A **transcript** says forty minutes and the timer is set to 40.

Name the first mismatch, propose a confirmation step and explain why the **transcript** cannot be treated as a complete copy of the recording.

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Sound Becomes a Signal

> A fictional microphone display records five samples during a short classroom tap. Explain the labelled step before checking the changed condition.

A fictional microphone display records five **sample**s during a short classroom tap.

Times are 0,1,2,3,4 ms in order.

Signal A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

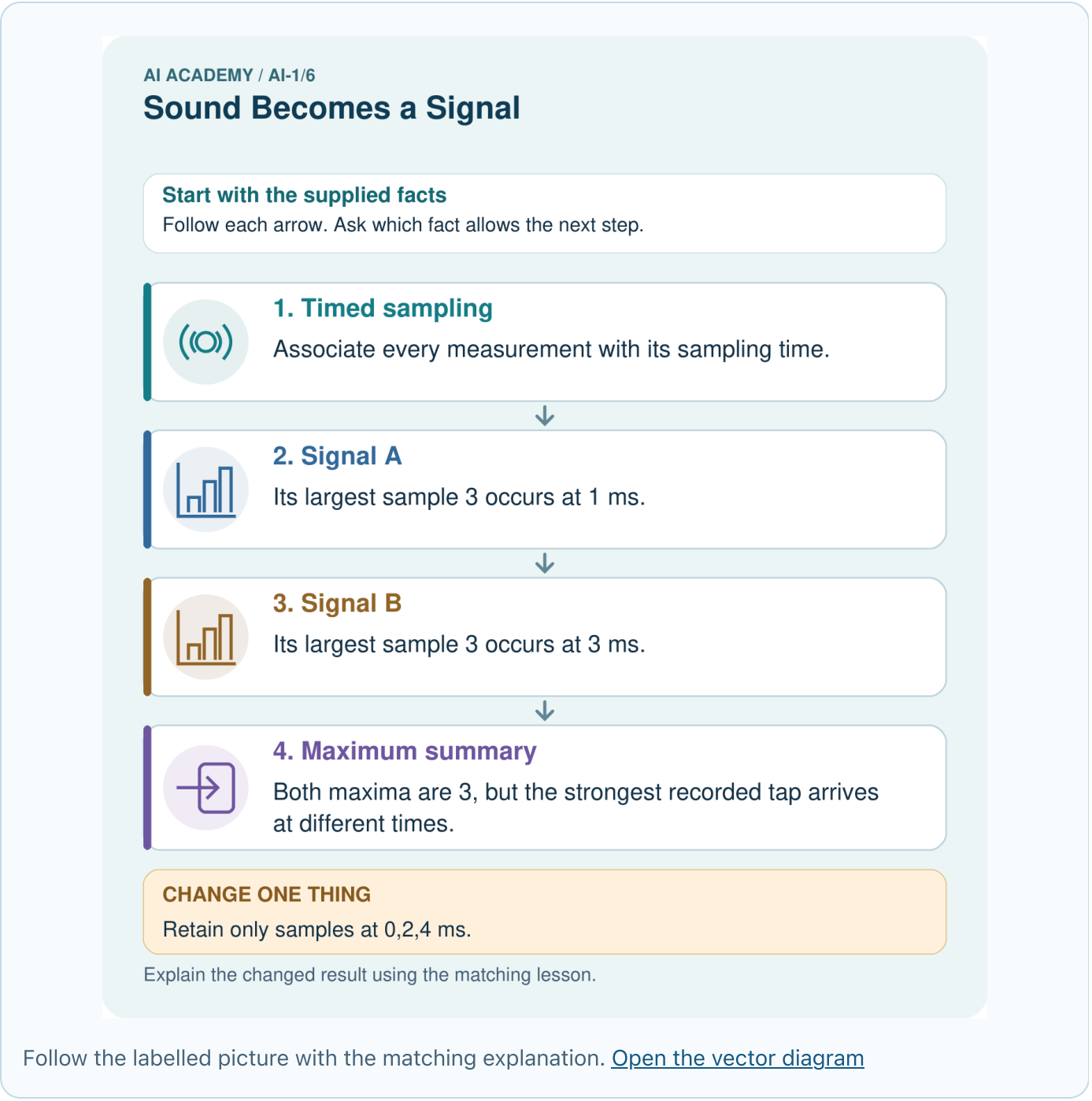
AI ACADEMY / AI-1/6

Same peak, different sample timing

	0 ms	1 ms	2 ms	3 ms	4 ms
Signal A	0	3	0	1	0
Signal B	0	1	0	3	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Does the all-zero retained record prove the full **signal** was silent? Use the picture.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional microphone display records five **sample**s during a short classroom tap.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both maxima are 3, but the strongest recorded tap arrives at different times.

1. **Signal** A

2. **Signal** B

3. Maximum summary

4. Timed sampling

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Maximum summary Both maxima are 3, but the strongest recorded tap arrives at different times. The sequences contain the same values in different orders.

Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Retain only **sample**s at 0,2,4 ms. Which conclusion is supported by

the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Both retained sequences are 0,0,0.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both retained sequences are 0,0,0. The selected times omit both nonzero **sample**s from each **signal**.

Repair a missing building block

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

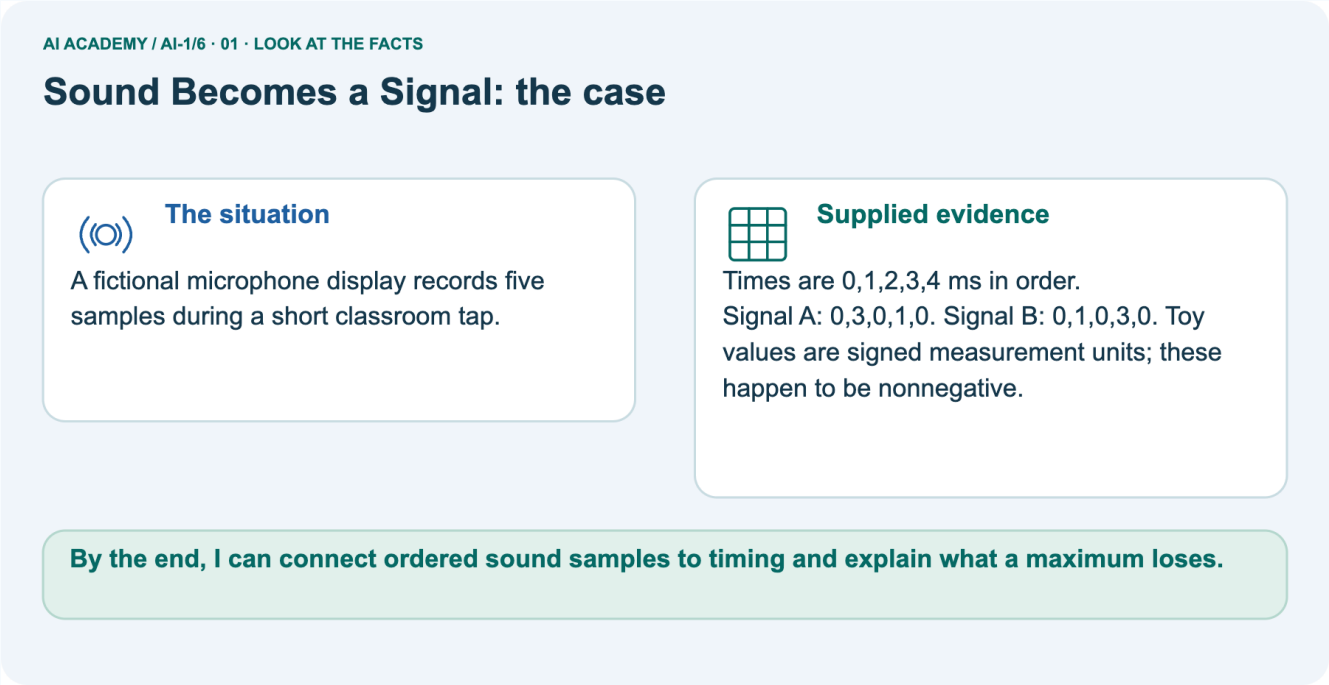
> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Sound Becomes a Signal picture from its labels, then record a relevant limit.

01. Sound Becomes a Signal: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

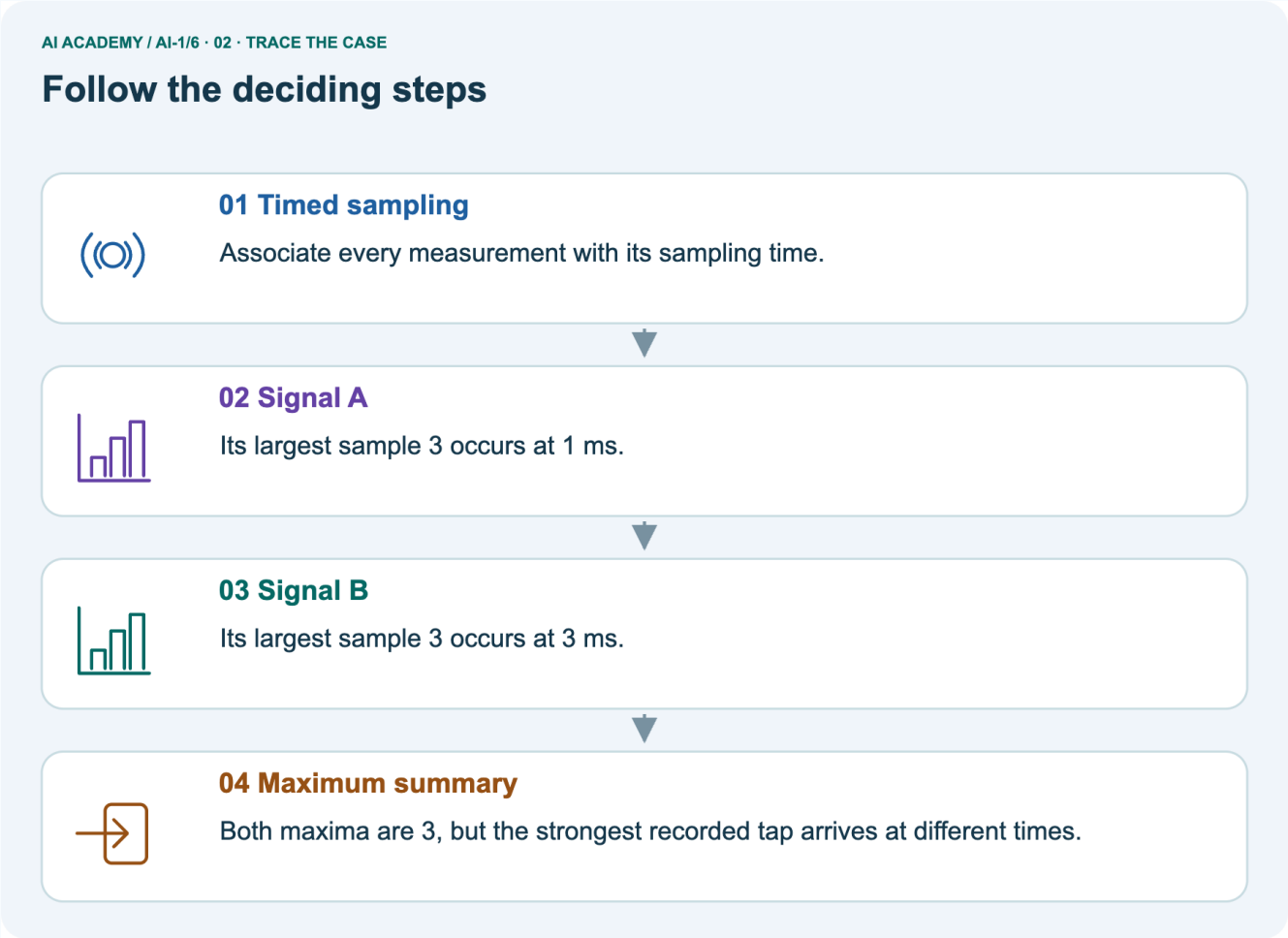
A fictional microphone display records five **sample**s during a short classroom tap. Supplied facts: Times are 0,1,2,3,4 ms in order. **Signal** A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0.

Toy values are signed measurement units; these happen to be nonnegative.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



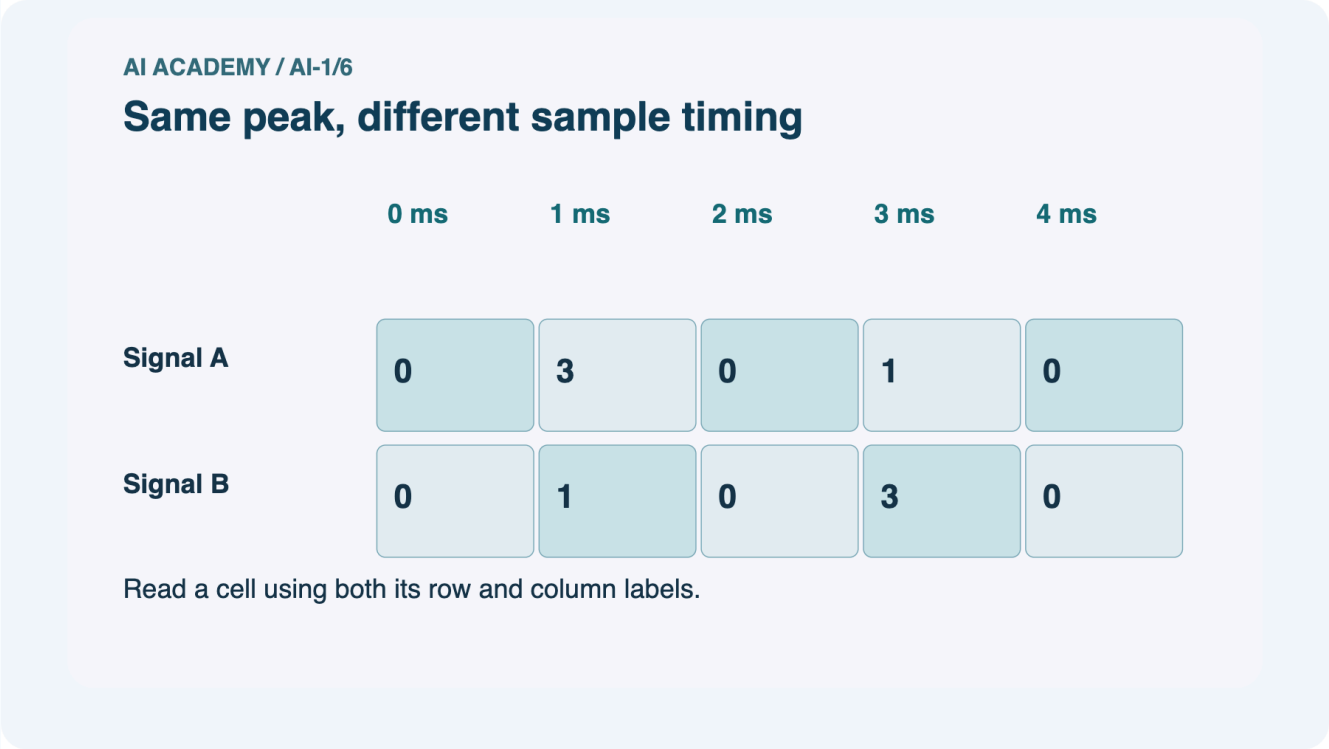
Follow the labelled picture with the matching explanation. Open the vector diagram

Timed sampling: Associate every measurement with its sampling time. → **Signal** A: Its largest **sample** 3 occurs at 1 ms. → **Signal** B: Its largest **sample** 3 occurs at 3 ms. → Maximum summary: Both maxima are 3, but the strongest recorded tap arrives at different times.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same peak, different sample timing



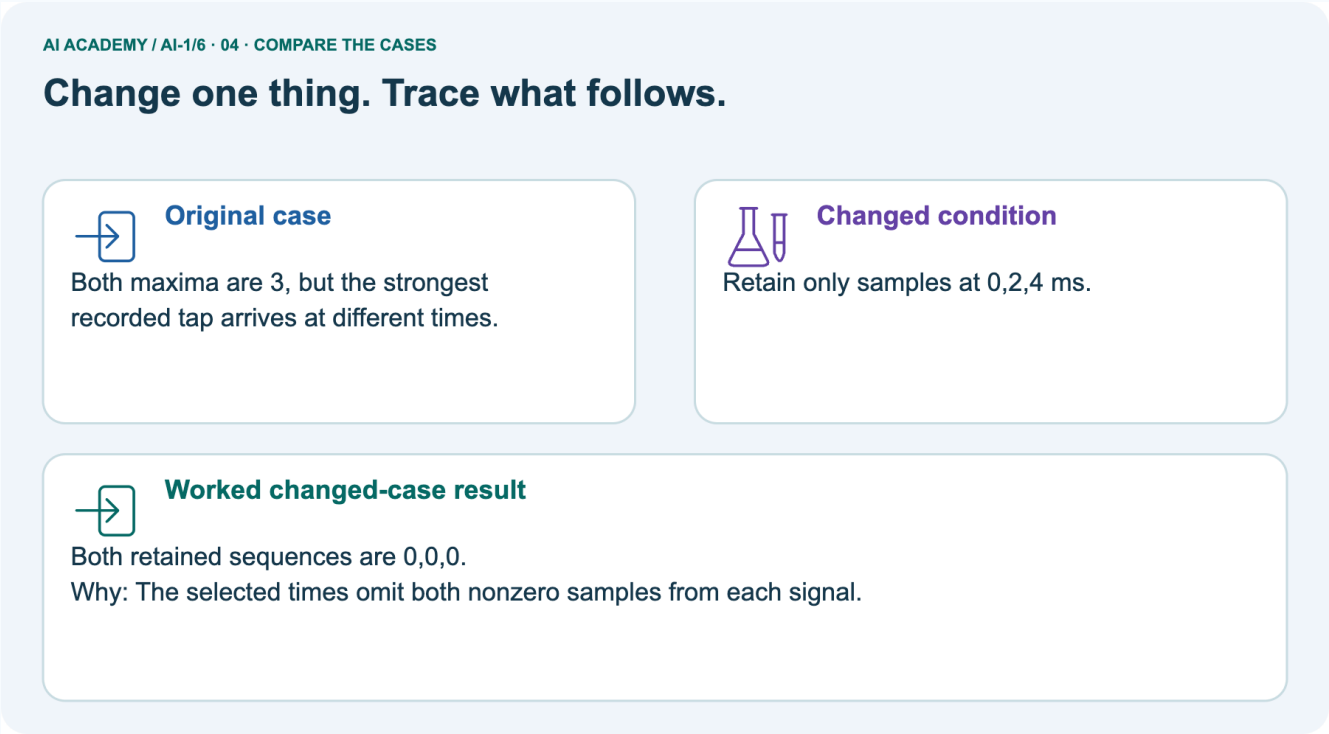
Follow the labelled picture with the matching explanation. Open the vector diagram

Same peak, different **sample** timing. **Signal** A: 0 ms = 0, 1 ms = 3, 2 ms = 0, 3 ms = 1, 4 ms = 0. **Signal** B: 0 ms = 0, 1 ms = 1, 2 ms = 0, 3 ms = 3, 4 ms = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

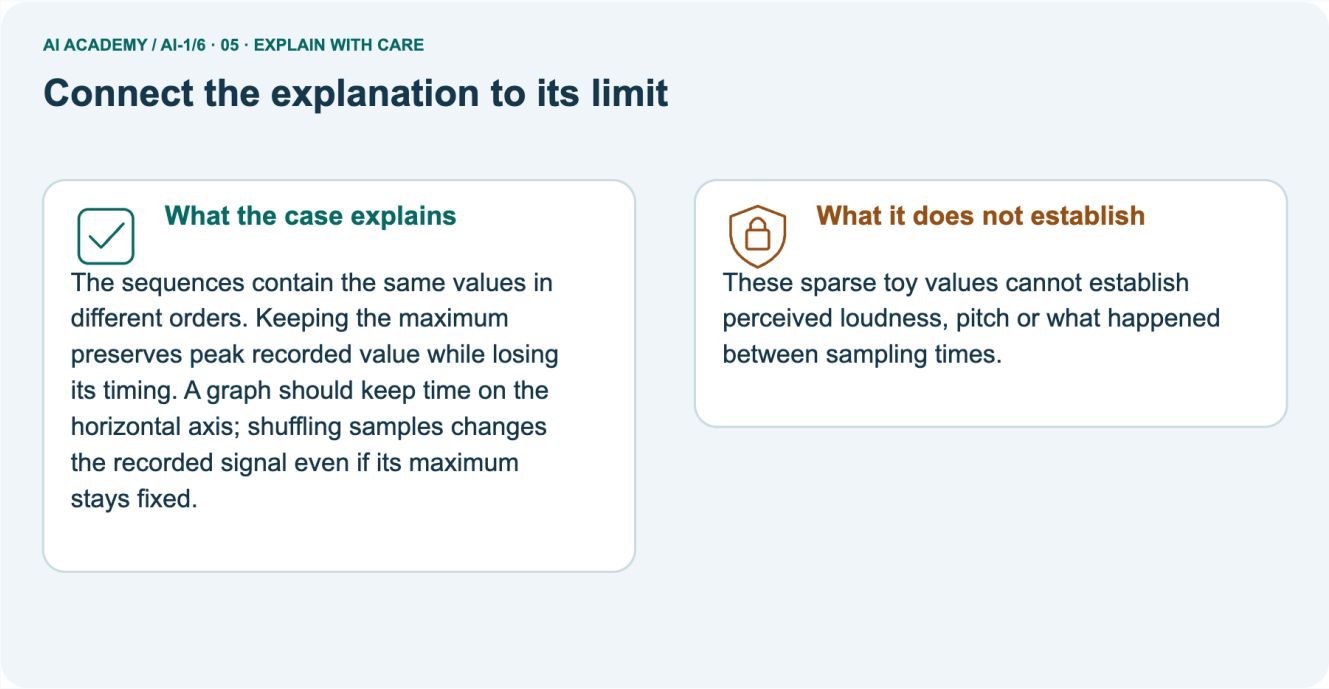
Original case: Both maxima are 3, but the strongest recorded tap arrives at different times. Changed condition: Retain only **sample**s at 0,2,4 ms. Expected result: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero **sample**s from each **signal**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

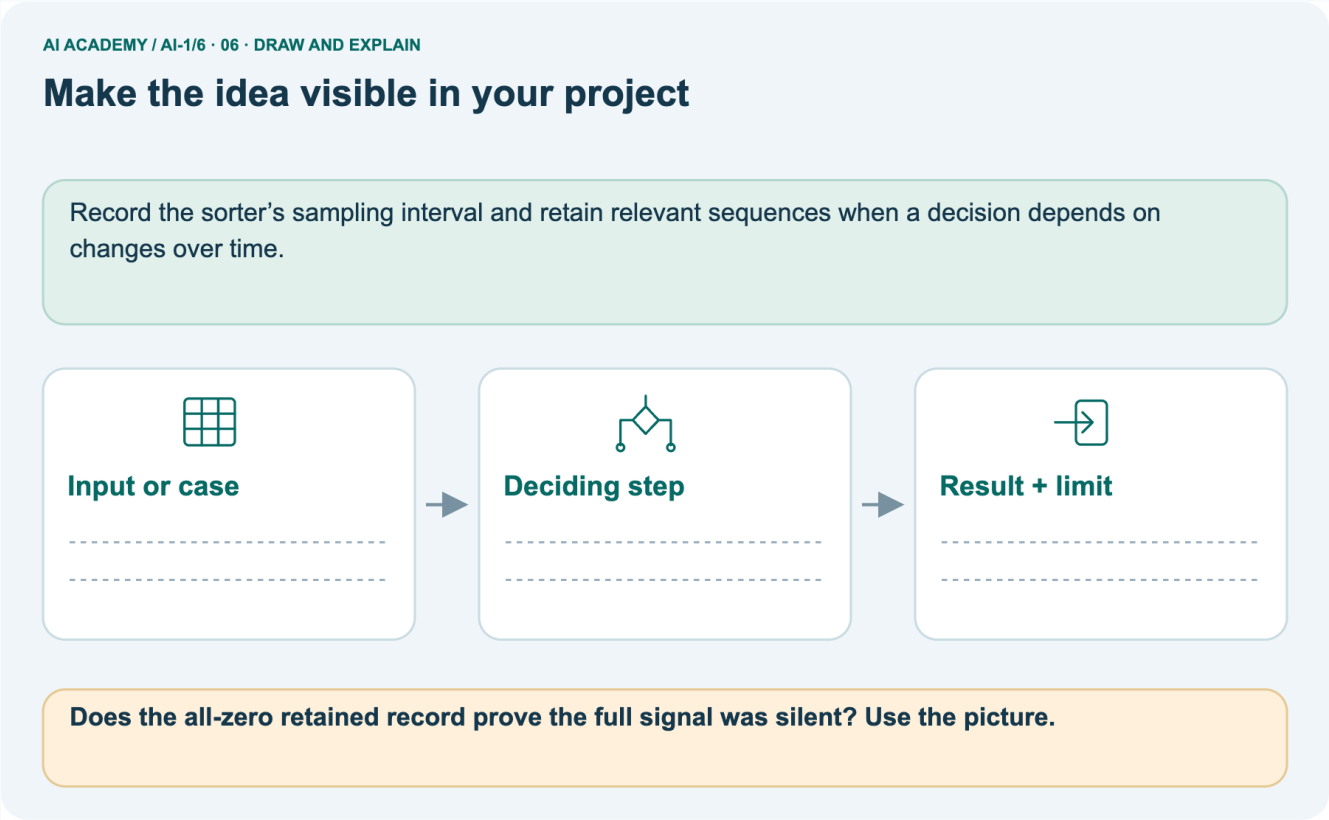
A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Limit: These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time. Draw your **input**, deciding step, result and limit.

Supported question: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.